

RFID SOLUTIONS FOR THE EXPRESS PARCEL AND AIRLINE BAGGAGE INDUSTRY

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There are currently many Radio Frequency Identification (RFID) solutions servicing wide diversity of applications that use both Active and Passive transponders. Over the years a range of products have been developed using different frequencies including, Low (125 – 134 kHz), Very High (314 MHz ~ 915 MHz) and Ultra High (2.45 GHz) frequencies. Each frequency offers to the user a unique set of features, e.g. cost, reading range, data speed and penetration but often confined to a country or geographic region by regulatory limitations. Many Companies have tried to develop the “holy grail” of RFID, where a passive tag loaded on the back of a wagon, is able to be read at distances exceeding 10 metres, whilst travelling at 50 kph through the factory gate and yet, costing only pence to purchase. Unfortunately, we are still along way off this utopian ideal as some of the features needed are mutually exclusive (and in turn would need to defy the laws of physics).

Most RFID solutions to date have been bespoke or turn key, using specially tailored solutions to overcome a unique problem. Typical applications in question include beer barrel tracking, warehousing, security access, vehicle immobiliser, car park entry tags, priority traffic light schemes, animal tagging and many more. All of these applications have used passive tags whose form factor was developed to fit the application e.g. laminated into paper or plastic labels, cards or rods. Applications using active tags are again a niche market, where these expensive tags are used when longer reading ranges are required. Typically these applications, where the transponder is constantly powered by a battery or connected to the vehicle electronics supply, are motorway tolling and container tagging.

For the past 3 years Texas Instruments has been developing a passive tag which removes some of these difficulties, meeting the optimum cost / performance trade off and performance requirements for two dedicated market sectors. In the form of a “Smart Label” the transponder electronics have been laminated into paper labels and plastic cards and this product is called **Tag-It™**.

This passive transponder has been developed for the target markets of Express Parcels and Airline Baggage though, over time, we expect that it will be adapted for use in other market sectors. In these two markets a flexible label is required where it can meet existing label specifications and form factors and be easily introduced into existing hardware.

Key to this development work were the following facts:

1. Use of the Industrial, Scientific and Medical (ISM) frequency band,
2. This frequency produces a low cost design for both the transponder and Readers
3. It allows high data rates
4. Creates a small foot print
5. Satisfies performance criteria required in these markets.
6. Migration path to different transponder configurations.

The 13.56 MHz carrier frequency chosen for this product has world-wide acceptance, which allows global usage of the technology. Equally important the associated reader hardware had to be competitive with other solutions on offer. This frequency allows the transponder to be small and flexible together with a high level of integration in the reader electronics, giving a small footprint, and allowing it to be readily integrated into hand held scanners. Designs also exist that allow larger read gate antenna arrays to be built for conveyor belt applications in warehouses or airports.

An important consideration in a global market is the need to adapt to the market requirements. Although the transponder operation will remain the same, its memory configuration could be different for each new market and this is easily achieved using the current integrated circuit technology.

How does it work? Simplistically, the passive transponder requires power in order for it to operate. This power is created by the Radio signal, with energy transferred as the transponder antenna passes through the magnetic field created by the antenna, generating a voltage across the charge pump circuit. This charge pump increases the voltage enabling the silicon chip to operate. Once the chip is charged and operating, the reader transmitter then sends the relevant commands to the transponder, which in turn returns its data to the receiver. This cycle is repeated many times a second.

The Transponder

The 13.56 MHz transponder antenna is made from etched aluminium on a Polyethyleneterephthalate (PET) substrate giving two footprints of 45 mm x 45 mm and 45 mm x 76 mm. The silicon chip is "bonded" onto the aluminium antenna making a very robust mechanical and electrical connection. The relative thickness of the overall inlay, at the chip, is 375µm, making it a very thin transponder inlay, which can be laminated within paper labels and plastic cards. The transponder inlay is supplied on reels so that laminators can fully automate the process of introducing the inlay into the paper label or plastic card.

Each silicon chip used in the transponder has the following elements, a charge pump circuit, decode logic, factory, and OEM programmable memory areas. The factory memory area is programmed at source with the manufacturer's code, a unique chip identification number (SID), the chip version and the memory configuration. Once these have been programmed into the memory area then the blocks are factory locked creating a read only zone. The present OEM memory area has 256 bits configured in 8 blocks of 32 bits where data can be stored. These blocks can be written to and read from many times, or written to and the data locked making it read only (WORM). The OEM blocks can also be pre-programmed at the factory and locked if a client requires this service. Robust techniques are employed to ensure that all two-way communication between the tag and reader are sent and received without error.

All responses from the transponder are noise resistant FM signals (FSK) which give the system the added advantage of "signal capture". This is where the reader will lock onto the strongest transponder signal, but if many transponders are expected then the Simultaneous Identification (SID) mode can be activated. In this mode any number of transponders can be identified, the number only being limited by the time that the transponders are able to remain within the RF field.

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The transponder is characterised for operating over the temperature range of -25° to $+70^{\circ}$ Centigrade and has a storage temperature of -40° to $+80^{\circ}$ Centigrade.

The Reader

The Reader comprises an antenna and transceiver unit. The design of the transceiver unit is a typical superhetrodyne configuration with power RF stage operating at 13.56 MHz. The current version has an adjustable output power from 800 mW to 2 Watts but to comply with CE conformance and ETSI approval, the demonstration units are limited to 800 mW. Using an antenna measuring 9 cm by 19 cm, a read range of 22 cm is achieved.

The transmitter uses a pulse width coded 100% amplitude modulated carrier frequency of 13.56 MHz to power and send the commands to the transponder. However, just an RF field created by the transmitter is not sufficient for the transponder to reply. There is a protocol flow that the transponder has to receive before it will send back its data. The transponder must:

- be in the reader RF field,
- received the complete Downlink protocol correctly
- remain within the RF-field whilst sending back the information.

The transponder is a slave and is not allowed to respond until requested by its master, the reader. This is a very important point because there may be other transponders in the field, using the same frequency but for different applications, this ensures that only the transponders important to your application will ever respond.

It is important to note that throughout the transmit and receive cycles the data is checked for errors by using a powerful CCITT CRC check.

The transponder transmits back its data using Manchester encoded signals to make efficient use of the receive cycle. The receiver decodes the transponder signal and sends the resultant information onto a host computer via an RS232 serial port.

No Reader is complete unless there is an antenna structure. This is used to illuminate the transponder, to send it commands and receive its data back. Differing sizes of antennae and coupling arrangements can be designed in order to make a satisfactory system depending on the application. The antennae can be readily constructed from wire, tape or copper pipe to give inductances that are easily tuned to resonance with capacitance. The Quality (Q) factor of the antenna is important because if this value is too high then the receiver bandwidth is reduced and you will reduce the read range as a consequence because it is the signal sidebands which carry the transponder data. It is also important to note that the output impedance of the transmitter is nominally 50 ohms, so additional matching may be required. The use of

balun's in the matching circuits is recommended to create balanced systems and to help reduce the effects of common mode noise.

Antenna matching circuits can adopt different techniques. Capacitive transformers, ferrite transformers, Gamma or T matched methods have all been used - some are easier than others to build and tune. Work continues with the matching circuits, but the ones finding favour are where it is easiest to set up and maintain, e.g. Gamma and T matching circuits.

Other antenna techniques exist in order to increase the read range of the transponder, the first is to have separate transmit and receive antennae or by using rotating field structures. By utilising these different techniques it is possible to successfully read transponders across a 1.2 metre conveyor system, as proven by the airline baggage trials recently conducted in Heathrow Airport and in Express Parcels.

Whilst there is a need for high power readers for conveyor and long read range applications, there is also a need for handheld and close proximity readers. 13.56 MHz frequency allows a high level of integration in the reader electronics. Today reader transceiver modules exist that satisfy this market demand, some handheld scanners even have combined RFID and barcode scanners embedded in their housings.

Conclusion

Overall, **Tag-it™** (13.56 MHz) products have been proven to meet some of the most stringent application demands during recent trials in Airline and Express Parcel applications. The low cost, flexibility, and high performance demonstrated by the technology will enable other markets to exploit the benefits.

In view of the potential high volume (billions) of transponders required in the "Smart Label" market, an "Open Standard" is being drafted under the auspices of ISO 15693. This common protocol will allow other transponder vendors to enter the market which will be compatible with the installed base of Readers thus increasing the supply and helping to maintain "smart label" demand.